Yoochan Lee

Postdoctoral Researcher

Max Planck Institute for Security and Privacy, Bochum, Germany

(+82) 10-5264-4225 | yoochan.lee@mpi-sp.org | leeyoochan.github.io | LinkedIn

Research Interests

My research improves system security by advancing **exploit techniques** that reveal the true severity of software vulnerabilities. By developing principled methods to assess **exploitability**—determining whether and how a bug can be weaponized—I enable defenders to accurately **prioritize remediation** and allocate resources where they matter most. My work spans **kernel exploitation**, **side-channel–assisted attacks**, and **data-oriented programming**.

Education & Career

Max Planck Institute for Security and Privacy — Bochum, Germany
Postdoctoral Researcher — *Nov 2025 – Present*
Advisor: Prof. Thorsten Holz

Seoul National University — Seoul, South Korea
M.S./Ph.D. in Electrical and Computer Engineering — *Sep 2019 – Aug 2025*
Advisor: Prof. Byoungyoung Lee

Arizona State University — AZ, USA
Visiting Scholar — *Mar 2024 – Jun 2024*

Hanyang University — Seoul, South Korea
B.S. in Computer Science and Engineering — *Mar 2012 – Feb 2018*

Publications

Peer-Reviewed

6. **GHost in the SHELL: A GPU-to-Host Memory Attack and Its Mitigation**
   Sihyun Roh, Woohyuk Choi, Jaeyoung Chung, **Yoochan Lee**, Suhwan Song, and Byoungyoung Lee
   *In IEEE Symposium on Security and Privacy (**S**&**P**), May 2026*

5. **DirtyFree: Simplified Data-Oriented Programming in the Linux Kernel**
   Yoochan Lee, Hyuk Kwon, and Thorsten Holz
   *In Network and Distributed System Security Symposium (**NDSS**), Feb 2026*

4. **PeTAL: Ensuring Access Control Integrity against Data-only Attacks on Linux**
   Juhee Kim, Jinbum Park, **Yoochan Lee**, Chengyu Song, Taesoo Kim, and Byoungyoung Lee
   *In ACM Conference on Computer and Communications Security (**CCS**), Oct 2024*

3. **Pspray: Timing Side-Channel based Linux Kernel Heap Exploitation Technique**
   Yoochan Lee, Jinhan Kwak, Junesoo Kang, Yuseok Jeon, and Byoungyoung Lee
   *In USENIX Security Symposium (**SEC**), Aug 2023*

2. **Diagnosing Kernel Concurrency Failures with AITIA**
   Dae R. Jeong, Minkyu Jung, **Yoochan Lee**, Byoungyoung Lee, Insik Shin, and Youngjin Kwon
   *In European Conference on Computer Systems (**EuroSys**), May 2023*

1. **ExpRace: Exploiting Kernel Races through Raising Interrupts**
   Yoochan Lee, Changwoo Min, and Byoungyoung Lee
   *In USENIX Security Symposium (**SEC**), Aug 2021*

Under Submission

- **Know Your Heap Before You Leap: Blind Reconnaissance for Environment-Agnostic Kernel Exploitation**
  Yoochan Lee, and Thorsten Holz
  *Submitted to USENIX Security Symposium (**SEC**), 2026*

- **Heap Localization: Cache Side-Channel based Linux Kernel Heap Exploit Techniques**
  Yoochan Lee, Sihyun Roh, Hyuk Kwon, Byoungyoung Lee, and Thorsten Holz
  *Submitted to IEEE Symposium on Security and Privacy (**S**&**P**), 2026*

Industrial Conferences

- **Privilege Escalation Exploit using DOP in x86-64 macOS**
  Yoochan Lee, Sangjun Song, Junoh Lee, and Jeongsu Choi
  Hack In The Box Amsterdam 2023

- **Perfect Spray: A Journey From Finding a New Type of Logical Flaw at Linux Kernel To Developing a New Heap Exploitation Technique**
  Yoochan Lee, Jinhan Kwak, Junesoo Kang, Yuseok Jeon, and Byoungyoung Lee
  BlackHat Europe 2022

- **Exploiting Kernel Races through Taming Thread Interleaving**
  Yoochan Lee, Changwoo Min, and Byoungyoung Lee
  BlackHat USA 2020

Teaching Experience

Teaching Assistant

Programming Methodology (430.211) — Spring 2025
Seoul National University

Cyber Security and Blockchain (M2177.006300) — Spring 2020, Fall 2020, Fall 2021, Fall 2022
Seoul National University

Mentoring

Best of the Best (BoB) — Mentor — Jul 2023 – Feb 2026
Government-funded cybersecurity education program (South Korea)

Mentored top-tier security students on kernel exploitation; guided responsible vulnerability disclosure (e.g., Linux, NVIDIA).

White Hat School — Lead Mentor — Sep 2023 – Dec 2025
Government-funded cybersecurity education program (South Korea)

Designed and delivered a system security curriculum (lectures and hands-on labs) for beginner-to-intermediate students.

Honors and Awards (Selected)

- **3rd Place**, DEFCON 30 CTF (Team StarBugs), Las Vegas, USA, Aug 2022
- **4th Place**, DEFCON 29 CTF (Team StarBugs), Las Vegas, USA, Aug 2021
- 11th Place, DEFCON 28 CTF (Team Star-Bugs), Las Vegas, USA, Aug 2020
- 15th Place, DEFCON 27 CTF (Team CGC), Las Vegas, USA, Aug 2019
- 1st Place, Cyber Conflict Exercise and Contest 2018 (GYG), Jeju, South Korea, Oct 2018
- 13th Place, DEFCON 26 CTF (Team C.G.K.S), Las Vegas, USA, Aug 2018
- Encouragement Award, Samsung CTF 2017, Seoul, South Korea, Aug 2017

- 9th Place, DEFCON 25 CTF (Team RRR), Las Vegas, USA, Aug 2017
- 1st Place, Secuinside Capture The Bug (Team Minionz), Seoul, South Korea, July 2016
- **Top 10**, Best Of the Best 4th Generation, Mar 2016

Industry Experience

Raon WhiteHat, Seoul, South Korea
Security Intern: Penetration Testing
Feb 2017 – Aug 2017

Naver Labs, Gyeonggi-do, South Korea
Security Intern: Browser Vulnerability Research (Naver Whale)
Apr 2016 – Jun 2016

ETRI, Daejeon, South Korea
Intern, Network Security Team
Jan 2015 – Feb 2015

Vulnerability Discoveries (Selected)

CVE-2021-31077 (macOS): Kernel heap overflow leading to Local Privilege Escalation.

Solidly Smart Contract: Critical vulnerability allowing unauthorized fund withdrawal (Tremendous funds drained).

CVE-2018-4417 (macOS): Kernel Information Leakage.

CVE-2018-4338 (macOS): Kernel Information Leakage.

CVE-2018-4084 (macOS): Kernel Information Leakage.

CVE-2017-7014 (macOS): Arbitrary Kernel Code Execution.